



Title: RESTAURANT MANAGEMENT SYSTEM

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Course Title : Programming Language II Lab

Department : CSE

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INTRODUCTION

1. Project Objective:

Develop a Java desktop application to simulate daily restaurant operations, allowing staff to add menu items, build customer orders, and track total sales—while demonstrating core OOP principles via a Swing-based GUI.

2. Project Overview:

The system lets a cashier/waiter add food items (Main Course or Dessert) with prices and optional service charges, build orders interactively, submit them for handling by a waiter and chef, and record total sales. It uses a user-friendly Swing interface and models real-world entities like Restaurant, Order, FoodItem, Waiter, and Chef

3.System Theory (OPP Concepts):

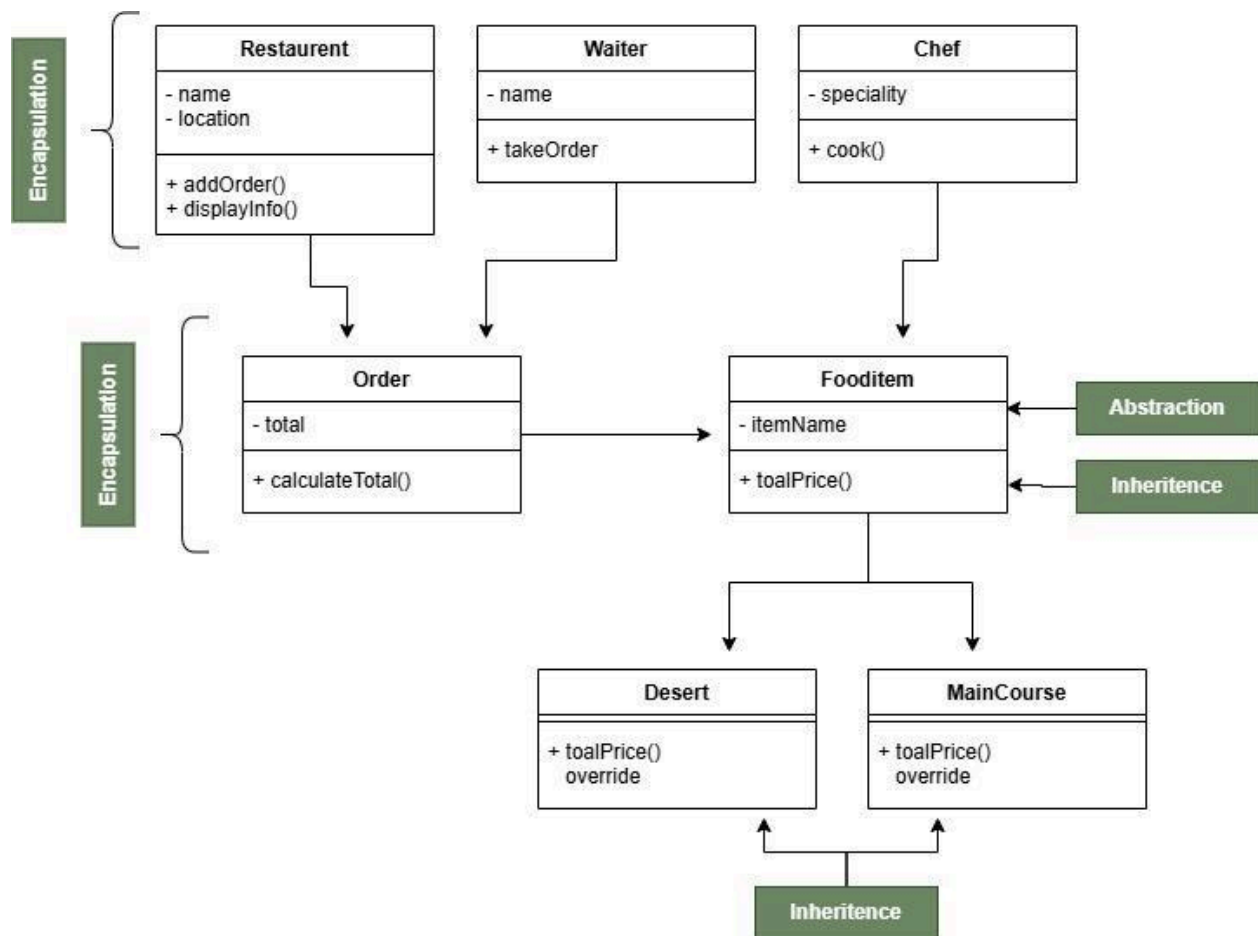
PRINCIPLE	IMPLEMENTATION
Encapsulation	Private fields (e.g., totalSales, orderId, itemPrice) with public getters/controlled methods.
Abstraction	Fooditem abstract class with abstract method totalPrice() – implementation hidden in subclasses.
Inheritance	MainCourse and Dessert extend Fooditem, reusing fields and getters.
Polymorphism	Both subclasses override totalPrice(). Same method call works for either type at runtime.

4.Object Model:

Class	Description
RestaurantGUI	Main driver with main() and Swing window; handles all user events.

Fooditem(Abstract)	Base class with itemName, itemPrice; declares abstract totalPrice().
Main Course	Extends Fooditem; implements totalPrice() returning the stored price.
Dessert	Extends Fooditem; implements totalPrice() returning stored price.
Order	Stores orderId and total; addItem(double price) accumulates bill.
Restaurant	Holds name, location, totalSales; addOrder(double amount) updates sales.
Chef	Stores name and specialty; cook() prints cooking message.
Test class	AbstractionTest, Nila, Marzia, Drive – separate test files, not part of main GUI.

5. Class Diagram:



6.GUI Screenshot:

Java Bites — Restaurant System

Java Bites Dhaka, Bangladesh | Waiter: Rahim | Chef: Karim

Add Food Item

Item Type: Main Course

Item Name:

Price (\$):

Service Charge (\$):

+ Add Item

✓ Submit Order

✗ Clear Order

Current Order

No items yet. Add items from the left panel.

Total: \$0.00

System ready. Welcome to Java Bites!

7.Conclusion:

The Restaurant Management System delivers a working food-ordering GUI using Java and Swing. It fully applies encapsulation, inheritance, polymorphism, and abstraction, separates GUI from logic, supports real-time order building, simulates staff roles, and tracks sales within a session.

Github Repository Link:

<https://github.com/Ro453/Restaurent-Mangement-System>